Battery Systems Engineer and Researcher
PhD candidate at TUM with 5+ years of applied research, specializing in data-driven modeling, state estimation, and optimization of grid-scale energy storage. Developed digital twin solutions for second-life battery systems through publicly funded, industry-collaborative research, resulting in 7 peer-reviewed publications (3 as first author).

MARTIN CORNEJO

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Technical Skills

Programming:
Python, Julia, MATLAB

Libraries:
pyomo, pandapower, JuMP, ModelingToolkit, SciML ecosystem

Development:
SimSES (Python), RecursiveGPs.jl (Julia)

Tools:
Git, Docker, Linux, GitHub CI

Languages

Spanish – Native
English – C1
German – C1
Italian – B1
Armenian – A2

Experience

Research associate 03/2021 – Present
TUM – Chair of Electrical Energy Storage Technology

- Developed data-driven models and digital twin frameworks for second-life battery storage systems in the publicly funded KI-M-Bat project, in collaboration with Kempten University of Applied Sciences, Fenecon, and STABL Energy.
- Team lead (6 members) for Application and System Integration: coordinated research planning and contributed to acquisition of multiple publicly funded projects.
- Supervised 37 student works (20 projects, 11 Master's and 6 Bachelor's theses); conducted tutorials for two lectures and one laboratory course.

Working student 03/2019 – 11/2020
Siemens Corporate Technology – Energy System Modelling Group

- Data acquisition, preprocessing, and analysis supporting multiple energy system models deployed in industry projects.
- Applied machine learning and image recognition techniques in a collaboration with the Siemens AI Lab to detect wind turbines in satellite imagery.

Education

Doctor of Engineering (in progress) 03/2021 – Present
TUM – Chair of Electrical Energy Storage Technology
Thesis: Field-Data-Driven Digital Twins for Modeling and Optimization of Second-Life Battery Storage Systems.

Master of Science Electrical Engineering and Information Technology 04/2018 – 01/2021
Technical University of Munich
Grade: 1,6 (Passed with Merit)
Master's Thesis: A decentralized optimal power flow method for real-time applications in power distribution systems.

Bachelor of Science Electrical Engineering and Information Technology 10/2014 – 03/2018
Technical University of Munich
Grade: 2,3 (Passed with Merit)
Bachelor's Thesis: Cost analysis of solar air conditioning systems.